INCH-POUND
MIL-DTL-24643/46G
27 November 2023
SUPERSEDING
MIL-DTL-24643/46F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LS2WAU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight)

First	-	Copper conductor, tin-coated, ASTM B286, 22 AWG with 7 strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.024 inch. Wall thickness may vary from nominal as necessary in order for the completed cable to meet the electrical requirements specified herein. Telephone identification code shall be applied by method 3.
Fourth	-	Two conductors cabled together with a nominal lay of 1½ inches to form a pair.
Fifth	-	The specified number of pairs (see table I) shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Braided shield of tin-coated copper.
Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.060 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:13Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/46	Type and size	Number of pairs	Overall diameter (inches)	
			min.	max.
-01UO	LS2WAU-40	40	1.320	1.370

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	17.71
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,500
Conductor to overall shield	2,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to overall shield	500
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	30
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C])	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:13Z
Check the source to verify that this is the current version before use.

Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:13Z
Check the source to verify that this is the current version before use.

Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS2WAU-40	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:13Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-062)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:13Z
Check the source to verify that this is the current version before use.